

HANHUI WANG

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EDUCATION

Ph.D. in Computer Science

Khoury College of Computer Sciences, Northeastern University (NEU)

Sept.2025 - Present

Advisor: **Prof. Huaizu Jiang**

M.S. in Computer Science

Viterbi School of Engineering, University of Southern California (USC)

Aug.2023 - May 2025

GPA 4.0/4.0

B.Eng. in Computer Science and Technology

Sept.2019 - June 2023

School of Computer Science & Technology (SCST)

Huazhong University of Science and Technology (HUST)

GPA 3.99/4.0 Rank 2/363

★ Selected by SCST for a special class for the most promising students

PUBLICATION & PREPRINT

1. **Hanhui Wang***, Yiming Xie*, Haiwen Feng, Zhaoyang Lv, Shenlong Wang[†], and Huaizu Jiang[†]. **Streaming Video Generation with Streaming Force Control**. In *Proceedings of the 19th ACM SIGGRAPH Conference and Exhibition on Computer Graphics and Interactive Techniques in Asia (SIGGRAPH Asia)*, 2026.
2. Aniket Gupta*, **Hanhui Wang***, Charles Saunders, Aruni RoyChowdhury, Hanumant Singh, and Huaizu Jiang. **SNAP: Towards Segmenting Anything in Any Point Cloud**. In *Proceedings of the 13th International Conference on 3D Vision (3DV)*, 2026, **Oral**.
3. Fangrui Zhu*, **Hanhui Wang***, Yiming Xie, Jing Gu, Tianye Ding, Jianwei Yang, and Huaizu Jiang. **Struct2D: A Perception-Guided Framework for Spatial Reasoning in MLLMs**. In *Proceedings of the 39th Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
4. **Hanhui Wang***, Yihua Zhang*, Ruizheng Bai, Yue Zhao, Sijia Liu, and Zhengzhong Tu. **Edit Away and My Face Will Not Stay: Personal Biometric Defense against Malicious Generative Editing**. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
5. **Hanhui Wang**, Huaize Ye, Yi Xia, and Xueyan Zhang. **Leveraging SAM for Single-Source Domain Generalization in Medical Image Segmentation**. In *arXiv 2401.02076*, 2024.

* Equal contribution. [†] Equal advising.

RESEARCH EXPERIENCE

Research Assistant

June 2024 - Aug.2025

Visual Intelligence Lab, Northeastern University

Advisor - **Prof. Huaizu Jiang**

- Introduced **SNAP**, the first unified segmentation model capable of handling diverse point cloud domains (indoor, outdoor, aerial) while supporting versatile prompt types such as spatial points (clicks) and text for flexible object segmentation.
- Introduced **Struct2D**, a perception-guided prompting framework that unlocks spatial reasoning in Multimodal Large Language Models (MLLMs) using structured 2D representations.

Research Assistant

June 2024 - Dec.2024

TACO Group, Texas A&M University

Advisor - **Prof. Zhengzhong Tu**

- Presented a novel perspective for protecting personal images from malicious editing, focusing on making biometric features unrecognizable post editing.

- Developed a new algorithm, **FACELOCK**, that incorporates facial recognition models and feature embedding penalties to effectively protect against diffusion-based image editing.
- Conducted a crucial analysis of the quantitative evaluation metrics commonly used in image editing tasks, exposing their vulnerabilities and highlighting the potential for manipulation to achieve deceptive results.

Research Assistant

Embedded and Pervasive Computing Lab, HUST

Sept.2022 - Mar.2023

Advisor - **Prof. Xianzhi Li**

- Modified a Few-shot learning framework for 3D Instance Segmentation (3DIS) to address the problem of the expensive costs of collecting a sufficient amount of annotated point clouds.
- Utilized a Transformer Decoder to generate differentiated kernels to perform instance-wise dynamic convolution.
- Implemented the model using **Python** and **PyTorch** and improved the mean Average Precision results (mAP) by **3.2** percent on the ScanNet V2 dataset.

WORKING EXPERIENCE

Assistant Algorithm Engineer

Research & Development Group (RDG), iFLYTEK

May 2023 - July 2023

- Worked on a 3D Instance Segmentation project aiming to combine the strengths of Clustering- and Transformer-based methods. Our model has achieved a result of **0.796** on the ScanNet V2 AP50 benchmark (higher than the previous state-of-the-art result of 0.787 by then).
- Worked on modifying the indoor scene instance segmentation model to improve performance on outdoor scene datasets.

COMMUNITY SERVICES

- Conference Reviewer: CVPR, CVPR Workshop, 3DV, ECCV, SIGGRAPH Asia
- Journal Reviewer: TOG

TALKS

- **SNAP: Towards Segmenting Anything in Any Point Cloud.** Feb. 2026
Invited talk at the Boston AI, ML, and Computer Vision Meetup, Microsoft Research New England (NERD).
- **Struct2D: A Perception-Guided Framework for Spatial Reasoning in MLLMs.**
 - Oral Presentation, New England Computer Vision (NECV) Workshop, UMass Amherst Nov. 2025
 - Spotlight Presentation, CVPR Workshop on Computer Vision in the Wild (CVinW), Nashville June 2025

TEACHING EXPERIENCE

- Teaching assistant of CS7150, Deep Learning, Northeastern University. Spring 2026

SELECTED AWARDS & HONORS

China National Scholarship (the highest national wide scholarship for undergraduate students in China)	2020
Outstanding Undergraduates in Term of Academic Performance (the greatest honor for undergraduates in HUST)	2020
Merit Student of HUST	2020, 2021, 2022
Outstanding Graduates of HUST	2023